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Therapeutic Landscapes in Rural Tourism: How Flow Experience, Nostalgia, and Restorative Experience Contribute to Spiritual Well-Being

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ABSTRACT

This study explores the role of therapeutic landscapes in rural tourism. While previous research has highlighted the relational nature of therapeutic landscapes, quantitative studies remain scarce, especially with respect to how therapeutic landscape experiences evolve dynamically. To address this gap, we propose a multiple mediation model that explores how flow experience, nostalgia, and restorative experience jointly influence the relationship between therapeutic landscape perception and spiritual well-being. Using three-wave survey data from 352 tourists visiting rural destinations in China, our findings confirm that therapeutic landscape perception positively influences spiritual well-being. Furthermore, this effect is mediated through two distinct pathways: a cognitive chain-mediating effect of flow experience and restorative experience and an emotional chain-mediating effect involving nostalgia and restorative experience. By expanding on existing research, this study enriches the understanding of therapeutic landscapes in rural tourism and offers valuable guidance for creating destinations that promote tourists' well-being.

1 | Introduction

Since the 1990s, therapeutic landscape research has been a crucial part of health geography (W. M. Gesler 1992; W. M. Gesler 1993; Taheri et al. 2021), and it gradually extended to tourism research (W. Gesler 1996; Huang and Xu 2018). Therapeutic landscapes are environments that help people improve physically, mentally, and spiritually (W. M. Gesler 1993; Moon 2020). Early studies viewed the healing properties of these landscapes as intrinsic (Bell et al. 2018; Moon 2020; Zhou, Cheng, and Rosenberg 2023). Recent research has increasingly recognized their relational nature. Conradson (2005) argued that therapeutic landscapes are not merely passive spaces; rather, they emerge through complex interactions between individuals and their broader socio-physical environment. This relational perspective has also influenced tourism research, particularly in regard to wellness tourism (Huang and Xu 2023; Yan and He 2020), health-driven seasonal tourism (Wang, Cui, and Xu 2024; Zhang et al. 2021), and therapeutic soundscapes (Kou and Xu 2024). However, much of this research has focused on specific populations, such as seasonal migrants (Wang, Cui, and Xu 2024; Zhang et al. 2021) or cancer patients (Huang and Xu 2023). Less attention has been paid to mass tourists. Rural destinations are increasingly being recognized as therapeutic landscapes (Kou and Xu 2024; Yan and He 2020). These environments may foster deep emotional and cognitive experiences that enhance tourists' overall well-being. Despite growing interest in this area, current research is still dominated by small-scale qualitative studies, and quantitative analyses still need to be further enriched (Zhou, Cheng, and Rosenberg 2023). In addition, how rural destinations shape tourists' therapeutic landscape

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experiences remains insufficiently explored. For example, what types of therapeutic experiences arise in these settings? How do different dimensions of these experiences interconnect? Addressing these questions is essential for developing a more comprehensive understanding of therapeutic landscapes in tourism research.

Research has consistently shown that well-being is a central outcome of engagement with therapeutic landscapes (Bell et al. 2015; Marques et al. 2021; Milligan and Bingley 2007). In tourism, the well-being of tourists is a growing concern (Karagöz and Ramkissoon 2024; Puhakka et al. 2017; Yu et al. 2021). Traditional rural tourism research mainly emphasized general well-being, focusing on physical relaxation, psychological restoration, and hedonic pleasure (Kou and Xu 2024). However, a growing body of literature suggests that well-being could extend beyond momentary happiness to encompass deeper dimensions (Meng et al. 2019). Spiritual well-being, which captures an individual's pursuit of meaning, purpose, and inner peace (Gedecho and Kim 2024; Han et al. 2024), offers exactly this deeper dimension. It represents how people assess their social, psychological, and subjective well-being (Gomez and Fisher 2003; Pan et al. 2021) and is a quality that people actively seek in tourism (Gedecho and Kim 2024; Li et al. 2021). Acknowledging spiritual well-being thus enables a more holistic understanding of rural therapeutic landscapes, moving beyond conventional frameworks of mere pleasure and escape. Furthermore, therapeutic landscape theory emphasizes relational interactions between people and places—not limited to physical elements but including mental and spiritual levels (Huang and Xu 2018). Given this relational perspective, exploring exactly how tourists' interactions with therapeutic landscapes enhance spiritual well-being becomes essential. This calls for a reassessment of how spiritual well-being is formed within the tourism therapeutic landscape.

Existing studies suggest that the therapeutic effects individuals experience in specific places may involve different mechanisms, including therapeutic assemblages (Foley 2011; Huang and Xu 2023), enabling places (Duff 2011; Zhou, Grady, and Rosenberg 2023), therapeutic mobility (Gatrell 2013; Chen and Wang 2022), and therapeutic taskscapes (Dunkley 2009; Smith 2021). These views have deepened our understanding of therapeutic landscape experiences from a relational perspective by emphasizing the dynamic interaction between place and individual. However, few studies have quantitatively investigated the dynamic transformation process of individuals' therapeutic landscape experiences. Bell et al. (2015) introduced a multidimensional model that unpacks how individuals experience therapeutic landscapes, identifying four interwoven dimensions: social, achieving, symbolic, and immersive. This model offers a compelling perspective for examining rural tourism's therapeutic potential and its impact on tourists' spiritual well-being. First, rural environments are often romanticized as spiritual hometowns (Meng et al. 2019). Here, individuals reconnect with ancestral heritage, embed themselves in family narratives, and immerse in cultural traditions that link the past to the present. This rekindled connection fosters self-reflection and social identity, mirroring the social dimension of therapeutic landscape experience. Second, rural destinations blend natural grandeur with cultural authenticity, offering a dynamic mix of recreational activities that induce flow experiences (King and Church 2013; Yang et al. 2023). These experiences, characterized by deep engagement and intrinsic enjoyment, correspond to the achieving dimension of therapeutic landscapes. Third, rural settings often carry strong nostalgic value, which acquires symbolic status as an idyllic alternative to urban environments (Yuan 2022). Tourists visiting these locations may recall connections to nature and traditional ways of living (Christou et al. 2018; Meng et al. 2019; Skuras et al. 2006), representing a symbolic therapeutic landscape experience. Fourth, rural destinations provide an escape from urban pressures, allowing tourists to immerse themselves in tranquil settings and simple lifestyles (Kou and Xu 2024; Yuan 2022). Such restoration aligns with the immersive dimension of therapeutic landscapes. Although Bell et al. (2015) multidimensional model provides a solid framework for examining therapeutic landscape experiences in rural tourism, current research has not sufficiently explored the linear relationships among these dimensions. Specifically, how do social therapeutic experiences—manifesting as spiritual well-being—develop in rural tourism? To answer this question, it is necessary to investigate the other dimensions of rural therapeutic landscape experience and uncover the mechanisms that connect them.

To address this gap, our study integrates Bell et al. (2015) therapeutic landscape experience model with Stienmetz et al.'s (2021) tourism experience creation framework. This integrated framework allows us to examine how therapeutic landscape perceptions influence tourists' spiritual well-being. While the therapeutic landscape experience model provides a foundation for understanding different types of therapeutic landscape experiences in rural tourism, the tourism experience creation model clarifies the relationships among these dimensions and their dynamic transformation processes. Specifically, we restructure the framework of rural therapeutic landscape experiences by identifying two core mechanisms: cognition (flow experience) and emotion (nostalgia). Furthermore, in alignment with the attention restoration viewpoint (Kaplan 1995), restorative experience serves as a chain mediator, linking the influence of therapeutic landscape perception to spiritual well-being through cognitive and emotional pathways.

2 | Literature Review and Hypothesis Development

2.1 | Theoretical Framework of the Study

The concept of therapeutic landscapes was first proposed by Wilbert Gesler, which has become a key theoretical contribution to health geography (Kearns and Milligan 2020; Kearns and Moon 2002). Therapeutic landscape theory explores the interplay between individuals and their physical, built, and social surroundings, shedding light on how these interactions foster wellness and enhance well-being (Bell et al. 2018; Huang and Xu 2018). Initially centered on healing sites, its application has broadened to include everyday environments (Bell et al. 2015; Zhou, Grady, and Rosenberg 2023) and tourism settings (Huang and Xu 2018; Zhang et al. 2021). Earlier research considered healing an intrinsic property of certain landscapes (Bell et al. 2018; Moon 2020; Zhou, Cheng, and Rosenberg 2023), but recent research highlights a relational perspective, suggesting therapeutic experiences are not predetermined but arise from

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interactions between individuals, society, and the physical environment (Conradson 2005; Huang and Xu 2023).

Expanding on this relational perspective, Bell et al. (2015) developed a model of therapeutic landscape experiences comprising four dimensions: achieving, symbolic, immersive, and social. The achieving dimension is linked to engaging in various activities, where physical challenges promote flow—an optimal state where skills match the difficulty of a task, fostering competence and fulfillment (Csikszentmihalyi 1990). In these moments, therapeutic spaces serve both functional and pleasurable purposes, facilitating personal growth and emotional release. The symbolic dimension pertains to places recognized for fostering health and well-being through cultural narratives (Bell et al. 2015; W. M. Gesler 1992). Personal symbolism, such as feelings of being "at home," further fosters comfort and strengthens emotional bonds (Bell et al. 2015). Immersive therapeutic experiences involve stepping away from daily routines, losing oneself, or gaining fresh perspectives through therapeutic encounters (Bell et al. 2015). This often entails attention restoration, emphasizing the embodied nature of renewal restorative encounters (Kaplan 1995; Wang et al. 2018). Engaging with a therapeutic environment's multisensory elements alleviates cognitive overload, clearing mental space for reflection (Herzog et al. 1997). The social dimension underscores the importance of relationships and context, encompassing friendly conversations, a joyful atmosphere, and bonding through shared activities (Bell et al. 2015). Here, a collective or "relational" sense of well-being emerges (Doughty 2013). Additionally, Bell et al. (2015) link these four dimensions to theoretical constructs such as flow, attention restoration, and peak diminutive experiences, providing a conceptual foundation for understanding how therapeutic landscapes enhance well-being.

Bell et al. (2015) model offers a valuable lens for analyzing rural therapeutic landscapes. Although studies have explored the healing potential of rural settings (Conradson 2005; Huang and Xu 2023; Yan and He 2020), few have considered this from the perspective of mass tourism. As rural areas combine unique local cultures and landscapes with diverse activity opportunities that facilitate tourists' flow, flow experiences may become a central component of achieving therapeutic experiences in these settings (Bell et al. 2015; Yang et al. 2023). Additionally, rural areas hold symbolic value as nostalgic alternatives to urban cities in cultural narratives (Yuan 2022), evoking encounters related to natural life and the past (Christou et al. 2018; Meng et al. 2019; Skuras et al. 2006). Hence, the nostalgic emotions of tourists are particularly significant in symbolic therapeutic experiences within rural contexts. Rural tourism also facilitates restorative experiences, providing an escape from fast-paced urban life (Kou and Xu 2024). Tourists immerse themselves in the slower rhythms of nature and traditional lifestyles (Christou et al. 2018), finding relief from daily stress (Kou and Xu 2024). Therefore, restorative experience can be understood as the immersive level of the therapeutic landscape experience in rural tourism. For some tourists, rural tourism holds deeper spiritual significance (Meng et al. 2019; Sharpley and Jepson 2011). The countryside represents a spiritual hometown (Meng et al. 2019), offering a way to reconnect with personal heritage, family stories, and cultural roots. This process contributes to spiritual well-being, which can be understood as an extension of social therapeutic experiences in a rural context. Therefore, on the basis of the therapeutic landscape experience model (Bell et al. 2015), this study aims to explore tourists' spiritual well-being in rural health tourism and to further investigate the roles of flow experience, nostalgia, and restorative experience within a multiple mediation model.

2.2 | Therapeutic Landscape Perception and Spiritual Well-Being

On the basis of therapeutic landscape theory, the intricate healing properties of therapeutic landscapes are revealed through their interaction with human subjects, indicating that the therapeutic effects of a landscape are shaped by how the landscape is perceived (Chen and Wang 2022). Therapeutic landscape perception, which involves individuals' embodied interactions with a place—physically, socially, and symbolically—plays an active role in the cocreation of landscape experiences (Menatti and da Rocha 2016; Zhang et al. 2021). The physical dimension refers to natural and built environments that offer health and therapeutic benefits (Zhang et al. 2021; Zhou, Cheng, and Rosenberg 2023). The social dimension refers to spaces shaped by social interactions, influenced by broader structures like hierarchy, race, and gender, as well as processes such as hegemony, ideology, and commercialization (Si et al. 2024; Zhang et al. 2021). The symbolic dimension involves symbolic interactions, including health beliefs, identity formation, and environmental meaning-making (Bell et al. 2018; Zhang et al. 2021; Zhou, Cheng, and Rosenberg 2023). This three-dimensional framework is widely used to analyze tourists' health perceptions (Zhang et al. 2021) and perceived quality of life at destinations (Si et al. 2024).

Well-being is closely linked to therapeutic landscapes (Brooke and Williams 2021; Marques et al. 2021). Rural destinations also have the possibility of providing spiritual experiences (Sharpley and Jepson 2011). Building upon these theoretical foundations, the relationship between therapeutic landscape perception and tourists' spiritual well-being warrants further exploration. Spiritual well-being, which represents the lived expression of an individual's spiritual health, is a vital part of one's overall quality of life and life satisfaction (Francis et al. 2015; Gedecho and Kim 2024; Li et al. 2021). Gomez and Fisher (2003) identified four ways to deepen our understanding of spiritual well-being: (1) the personal dimension, which focuses on life meaning, purpose, and direction (Li et al. 2021; Pan et al. 2021); (2) the communal aspect, which concerns the quality of relationships with others and is characterized by love, hope, and trust (Gomez and Fisher 2003); (3) the environmental dimension, which reflects harmony between human beings and is often accompanied by awe, wonder, and peace (Gomez and Fisher 2003); and (4) the transcendental aspect, which involves a deep connection with divine or transcendent forces (Li et al. 2021).

Rural therapeutic landscapes have the potential to enhance tourists' spiritual well-being through multiple pathways. Initially, traditional therapeutic landscapes view natural environments as prime spaces for well-being (Brooke and Williams 2021). Rural destinations offer abundant resources—such as natural landscapes, agricultural activities, and traditional architecture—that promote intimate contact with nature and rural life,

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enhancing tourist well-being (Kou and Xu 2024). Specifically, the diversity, grandeur, and vastness of nature evoke a sense of awe and humility, reminding tourists of their place in the larger natural world (Wang, Blasco, et al. 2024; Wang, Cui, and Xu 2024). This perspective highlights the role of the physical dimension of therapeutic landscape perception in fostering spiritual well-being.

Furthermore, participatory activities play a crucial role in promoting spiritual well-being (Gedecho and Kim 2024). In rural settings, staying in a farmhouse bed-and-breakfast with family-style hospitality, engaging in folk activities, and participating in outdoor sports enrich the tourism experience (King and Church 2013; Meng et al. 2019; Yang et al. 2023). These activities enable tourists to interact with local residents and fellow tourists (Meng et al. 2019; Wang, Blasco, et al. 2024; Wang, Cui, and Xu 2024). Removed from the pressures of modern life, social exchanges feel more organic, encouraging shared understanding and collective spiritual inspiration (Wang, Blasco, et al. 2024; Wang, Cui, and Xu 2024). Therefore, the social dimension of therapeutic landscapes actively nurtures spiritual well-being.

Beyond social and physical influences, a destination's symbolic value can contribute to spiritual healing (Bell et al. 2018). Research suggests that tourists often perceive rural areas as escapes from the noise, congestion, and stress of urban environments (Kou and Xu 2024), embracing them as places of health, authenticity, and nostalgia (Christou et al. 2018; Huang and Xu 2018; Meng et al. 2019). By ascribing symbolic significance to these landscapes, tourists unwind, reflect, and experience psychological relief (Jepson and Sharpley 2015; Kastenholz et al. 2020; Kou and Xu 2024), all of which contribute to spiritual well-being. Moreover, for some tourists, rural tourism is not merely a leisure activity; rather, it is a deeply personal journey influenced by traditional values such as the longing for an "old home" or the search for ancestral roots (Meng et al. 2019). Even in places with no direct familial ties, tourists may still cultivate a sense of belonging, reinforce personal identity, and fulfill deeper aspirations for self-actualization (Meng et al. 2019), further enriching their spiritual well-being. From the preceding discussion, we derive the following hypotheses:

H1. *Therapeutic landscape perception is positively associated with spiritual well-being.*

2.3 | Mediation of Restorative Experience

As previously noted, in rural tourism destinations, restorative experiences are an integral part of immersive therapeutic experiences in Bell et al. (2015) model. Restorative experiences refer to opportunities to reduce the fatigue of directed attention (Kaplan 1995; Kaplan and Kaplan 1989). The key elements of such experiences include coherence, compatibility, being away, and fascination (Kaplan 1995; Wöran and Arnberger 2012). Coherence refers to the perception that the environment is rich and aligns with an individual's cognitive understanding of it (Kaplan 1995; Kaplan and Kaplan 1989; Wöran and Arnberger 2012). Compatibility reflects the match between personal needs or preferences and the external environment (Kaplan 1995; Kaplan and Kaplan 1989). Being away involves temporarily escaping from reality, immersing oneself in an alternative world, and experiencing a deep sense of presence (Kaplan 1995; Rosenbaum et al. 2020). Fascination, the core of restorative experiences, occurs when the environment offers captivating elements that easily attract attention, fostering engagement (Kaplan 1995; Lehto 2012). Thus, therapeutic landscapes may enhance spiritual well-being through the following mechanisms: coherence, which brings order through structured environments; compatibility, which ensures alignment with personal values; being away, which relieves stress by immersing individuals in extraordinary settings; and fascination, which maintains engagement through captivating elements. These dimensions dynamically work together, fostering an ongoing recovery process that ultimately enhances tourists' spiritual well-being.

Rural tourism has increasingly been recognized for its restorative potential (Zhai et al. 2025). From a physical perspective, rural landscapes are particularly harmonious with nature (Tang et al. 2024). The natural aesthetics of rural farmland, including its textures, blue-green spaces, and fresh air, align with tourists' visual preferences and contribute to a restorative experience (Tang et al. 2024). On the social level, rural therapeutic landscapes also promote recovery. Engaging in activities such as walking in rural settings allows individuals to experience recovery (Doughty 2013; Roe and Aspinall 2011). Additionally, rural landscapes provide a neutral space for tourists to engage in social interactions beyond socioeconomic barriers, forming new connections and further contributing to restorative experiences (Zhang et al. 2021). Symbolic therapeutic landscapes further contribute by embodying cultural myths, sacred sites, or health-related symbols (W. M. Gesler 1993; Huang and Xu 2018), which support restorative experiences (Zhang et al. 2021). For example, tourists may meet centenarians to uncover longevity secrets, reinforcing the therapeutic value of destinations (Huang and Xu 2018). Studies consistently emphasize the role of restoration in individual well-being (Milligan and Bingley 2007; Qiu et al. 2022; Roe and Aspinall 2011). Research has also mentioned the significance of perceived destination restorative qualities for promoting spiritual health and fulfillment (Si et al. 2024). Thus, we hypothesize that, in rural tourism destinations, tourists' perceptions of therapeutic landscapes promote their spiritual well-being through restorative experiences, leading to the following hypothesis:

H2. *Restorative experience mediates the positive relationship between therapeutic landscape perception and spiritual well-being.*

2.4 | Multiple Mediation of Flow Experience and Nostalgia

Flow experience is a key component of achieving therapeutic experience, as defined by Bell et al. (2015). Previous studies have shown that the highly participatory interactions that tourists have with therapeutic landscapes set these environments apart from their daily lives and help them enter a state of liminality (Atkinson and Robson 2012; Smith 2021). This state closely resembles the flow experience, which occurs when individuals become fully immersed in an

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activity that aligns their skills with the challenges at hand (Csikszentmihalyi 1990; Jackson and Marsh 1996). In rural therapeutic landscapes, specific activities can engage tourists, facilitating flow experiences (Pitt 2014; Yang et al. 2023). As a result, therapeutic landscape perception in rural areas may be positively correlated with flow experience. Additionally, flow experiences are commonly linked to positive outcomes, such as enjoyment, satisfaction, and well-being (DeMatos et al. 2021; Yang et al. 2023). Tourism research indicates that flow-like experiences can serve as a source of spiritual nourishment for tourists (Andriotis 2011). Accordingly, within rural therapeutic landscapes, flow experience is likely to have a positive correlation with spiritual well-being. Building on this premise, we put forth the following hypotheses:

H3. *Flow experience mediates the positive relationship between therapeutic landscape perception and spiritual well-being.*

Nostalgia is integral to rural tourism and aligns with Bell et al. (2015) concept of symbolic therapeutic experience. It is often described as a bittersweet emotion, intertwines longing, sadness, and an irretrievable past with warmth, happiness, and meaningful memories (Newman et al. 2020; Sedikides and Wildschut 2016). Nostalgia can arise not only from personal past experiences but also from external triggers that evoke cherished moments (Chen et al. 2014; Meng et al. 2019). Rural therapeutic landscapes, rich in symbolism, provide fertile ground for nostalgia (Yuan 2022). Rural tourism activities enable visitors to ascribe personal symbolic meanings, which are often linked to past emotions and memories (Meng et al. 2019). For some tourists, activities such as picking fruit, exploring traditional farmhouses, or engaging in agricultural work recall childhood experiences or hometown lifestyle (Liang et al. 2021; Meng et al. 2019). Research consistently demonstrated the positive effects of nostalgia on well-being and the satisfaction of one's spiritual needs (Karagöz and Ramkissoon 2024; Meng et al. 2019).

Nostalgia aids in shaping environmental perception and strengthens life's continuity and meaning (Wood et al. 2015), which are necessary for spiritual well-being. These effects are particularly pronounced in rural settings, where tourists recall cherished moments, further deepening nostalgic emotions and enhancing overall well-being (Meng et al. 2019). Expanding on this, we propose:

H4. *Nostalgia mediates the positive relationship between therapeutic landscape perception and spiritual well-being.*

Although Bell et al. (2015) introduced a four-dimensional model of therapeutic landscape experiences, their approach was grounded in a nonlinear and networked perspective. While this perspective conceptualizes various experiences, it does not clarify the potential transmission pathways between different dimensions. In the field of tourism research, Stienmetz et al. (2021) proposed a framework of tourism experience creation, drawing on embodied cognition and emotional perspectives. This framework suggests that experience formation begins when environmental stimuli are processed through human sensory organs (Stienmetz et al. 2021). Subsequently, cognitive and emotional subsystems—including the tourist's brain, mind, and body—respond to the external environment, shaping the overall tourism experience (Stienmetz et al. 2021). Research has identified flow experience as a cognitive process (Csikszentmihalyi 1990; Simlesa et al. 2018; Yang et al. 2023) and nostalgia as an emotional process (Chen et al. 2014; Christou et al. 2018; Sedikides and Wildschut 2016).

Building on these insights, this study posits that flow experience and nostalgia act as mediators linking therapeutic landscape perception to spiritual well-being. From a cognitive perspective, rural tourism demands active and embodied engagement, fostering flow experiences (Bai et al. 2024; Csikszentmihalyi 1990; Yang et al. 2023). Such experiences allow tourists to break away from their everyday surroundings and routines (Pitt 2014). Moreover, flow experiences, which balance personal skills with environmental challenges (Csikszentmihalyi 1990; Jackson and Marsh 1996), lead tourists to perceive rural therapeutic landscapes as compatible settings. Therefore, flow experiences serve as the cognitive precursor to restorative experiences. On an emotional level, when rural landscapes intertwine with memories (Christou et al. 2018; Jepson and Sharpley 2015), nostalgia arises. This nostalgia leads tourists to be fascinated, creating a soft yet profound attraction to the landscape. The landscape, perceived through the lens of nostalgia, is coherent with their understanding (Meng et al. 2019). Consequently, nostalgia serves as an emotional precursor to restorative experiences. Previous research has also shown that restorative experiences foster benefits, including improved life quality (Hipp et al. 2016; Si et al. 2024), life satisfaction (Kawakubo and Oguchi 2019), and well-being (Qiu et al. 2022). Considering these findings alongside H2–H4, we further hypothesize:

H5. *Flow experience and restorative experience sequentially mediate the positive relationship between therapeutic landscape perception and spiritual well-being.*

H6. *Nostalgia and restorative experience sequentially mediate the positive relationship between therapeutic landscape perception and spiritual well-being.*

The theoretical model of this study is presented in Figure 1.

3 | Methods

3.1 | Participants and Procedures

This study adopted a convenience sampling strategy to recruit participants. Data were gathered from four rural tourism destinations—Nahui, Yancha, Xizhou, and Shuanglang Village—across Guizhou, Fujian, and Yunnan Provinces in China between February and July 2022. Tourists visiting these destinations on their first day and planning a stay of at least 2 days were invited to take part, yielding a final sample of 450 respondents. To boost participation, respondents received a small gift for completing the first questionnaire and an additional 5-yuan reward for the follow-up survey.

To reduce common method bias (CMB), we introduced a procedural safeguard by temporally separating the collection

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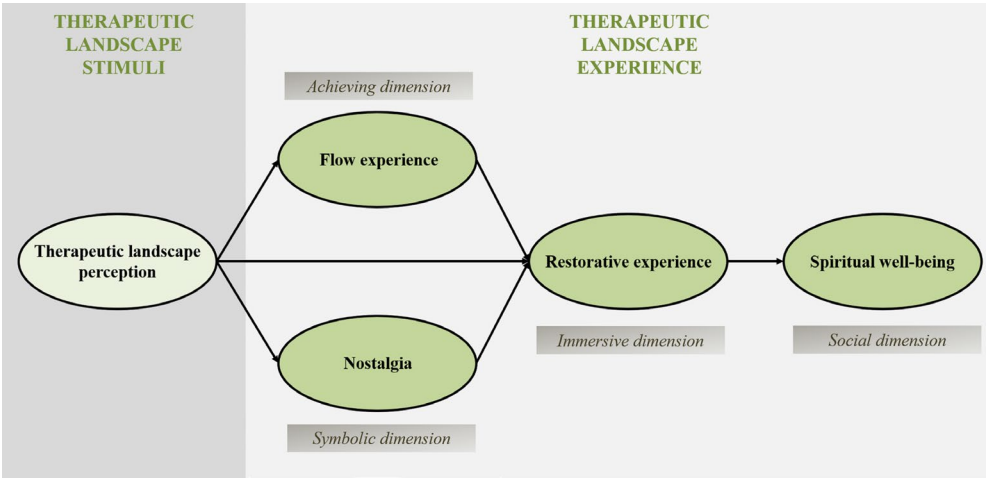


FIGURE 1 | Theoretical model.

of predictor and outcome variables (Podsakoff et al. 2003). Specifically, data were collected in three waves to align with the temporal characteristics of tourists' therapeutic experiences (Cui 2023; Hanel et al. 2024; Jian et al. 2021; Yang et al. 2023; Zhang et al. 2021). On the morning of DAY 1, we measured therapeutic landscape perception, flow experiences, and nostalgia. That evening, we assessed restorative experiences. On the night of DAY 2, we measured spiritual well-being and collected demographic information. Participants created unique IDs using the last four digits of their phone numbers to ensure accurate data matching across the three survey waves. Researchers sent electronic questionnaires and issued reminders at each stage.

After 98 incomplete or careless responses were excluded, 352 participants successfully completed all three surveys, resulting in a final response rate of 78.22%. In Wave 1, we distributed questionnaires to 450 volunteers and received 446 valid responses (114 from Nahui, 148 from Yancha, and 184 from Xizhou and Shuanglang Village). In Wave 2, among the 446 initial respondents, 370 provided valid responses (99 from Nahui, 107 from Yancha, and 164 from Xizhou and Shuanglang Village). In Wave 3, 352 participants completed the final survey (95 from Nahui, 98 from Yancha, and 159 from Xizhou and Shuanglang Village). The sample comprised 57.7% females and 42.3% males, with 43.8% aged 21–30. Most held an undergraduate degree (54.3%) and earned between 5000 and 8000 yuan per month (36.6%).

3.2 | Measures

Survey instruments were adapted from validated scales, with all non-demographic items measured on a 7-point Likert scale (1 = *strongly disagree*, 7 = *strongly agree*). As detailed in Appendix A, therapeutic landscape perception was measured using 12 items (Huang and Xu 2018; Wang et al. 2018; Zhang and Xu 2019; Zhang et al. 2021). Flow experience was assessed with a 3-item scale from Kim and Thapa (2018), which was contextually adapted from Yang et al. (2023). Nostalgia was measured using 7 items (Jian et al. 2021; Newman et al. 2020; Xu and Zhou 2020). Restorative experience was evaluated with a 12-item scale adapted from Kaplan (1995) and Lehto (2012). Spiritual well-being was also measured using 12 items (Gomez and Fisher 2003; Li et al. 2021; Pan et al. 2021). To rule out alternative explanations, demographic variables (i.e., gender, age, education level, and monthly income) were recorded and controlled (Jian et al. 2021; Zhang et al. 2021).

4 | Results

4.1 | Descriptive Statistics

SPSS 23.0 was utilized to test data normality. The skewness ranged from −1.20 to −0.30, and kurtosis values spanned from −0.65 to 1.33, affirming a well-distributed dataset (Kline 2011). Harman's single-factor analysis revealed that the first factor explained 41.87% of the variance, which is under the 50% cutoff, indicating that CMB is unlikely to be a major issue (Harman 1976; Podsakoff et al. 2003).

The composite reliability (CR) values for therapeutic landscape perception, flow experience, nostalgia, restorative experience, and spiritual well-being were above the cutoff value of 0.70 (Fornell and Larcker 1981). The average variance extracted (AVE) values for all variables were above the cutoff value of 0.50 (Fornell and Larcker 1981) (see Appendix B). Furthermore, the square root of the AVE for each construct exceeded its correlations with the other variables (see Table 1). Therefore, these results provide evidence of the convergent and discriminant validity of the five variables above.

The Pearson correlation results for the five variables provide preliminary evidence for the hypotheses. For example, therapeutic landscape perception is positively correlated with flow experience, nostalgia, restorative experience, and spiritual well-being ($r = 0.65$, 0.60, 0.70, and 0.53, respectively; $p < 0.01$). In addition, flow experience is positively correlated with restorative experience and spiritual well-being ($r = 0.67$ and 0.47, respectively; $p < 0.01$), whereas nostalgia is positively correlated with restorative experience and spiritual well-being ($r = 0.60$ and 0.63,

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respectively; $p < 0.01$). Furthermore, restorative experience is positively correlated with spiritual well-being ($r = 0.61$; $p < 0.01$).

4.2 | Confirmatory Factor Analyses

As shown in Table 2, the five-factor model achieved a good fit to the data ($\chi 2 = 2476.70$, df = 979, RMSEA = 0.06, CFI = 0.90, TLI = 0.90, SRMR = 0.05), and the fit index of the baseline five-factor model was better than that of the other models.

4.3 | Hypothesis Testing

First, the regression analysis confirmed a significant positive association between therapeutic landscape perception and spiritual well-being ($\beta = 0.81$, SE = 0.07, $p < 0.001$), supporting H1. Next, mediation analysis (see Table 3) using *Mplus* 7.0 and the bias-corrected bootstrap method (Muthén and Muthén 2012) further showed a significant indirect effect via restorative experience ($\beta = 0.20$, SE = 0.06, $p < 0.001$), with a 95% CI = [0.11, 0.33] excluding zero, supporting H2. Path analysis results indicated no significant positive association between flow experience and spiritual well-being ($\beta = -0.22$, SE = 0.32, $p = 0.48 > 0.05$). Furthermore, the indirect effects of therapeutic landscape perception on spiritual well-being through flow experience were not significant ($\beta = -0.05$, SE = 0.07, $p = 0.48 > 0.05$), and the 95% CI = [−0.18, 0.09] included zero with 5000 bootstrap samples. Thus, H3 is not supported. Conversely, the indirect effect through nostalgia was significant ($\beta = 0.37$, SE = 0.06, $p < 0.001$), with a 95% CI = [0.26, 0.50] excluding zero, confirming H4.

Figure 2 illustrates the multiple mediation analysis results. Sequential mediation analysis (see Table 3) further confirmed that therapeutic landscape perception impacted spiritual well-being through flow experience and restorative experience in sequence, with a significant indirect effect ($\beta = 0.10$, SE = 0.03, $p < 0.01$, 95% CI = [0.05, 0.18] did not include zero), validating H5. The same effect through nostalgia and restorative experience in turn was significant ($\beta = 0.06$, SE = 0.02, $p < 0.05$, 95% CI = [0.02, 0.12] did not include zero), which supports H6. The results further revealed that the direct effect of therapeutic landscape perception on spiritual well-being was nonsignificant ($\beta = 0.12$, SE = 0.10, $p = 0.22 > 0.05$), with the 95% CI = [−0.08, 0.31] excluding zero. These results present a full multiple mediation effect via flow experience, nostalgia, and restorative experience on the relationship between therapeutic landscape perception and spiritual well-being.

TABLE 1 | Results of the descriptive statistics.

	1	2	3	4	5
1. Therapeutic landscape perception	**0.73**				
2. Flow experience	0.65**	**0.82**			
3. Nostalgia	0.60**	0.57**	**0.81**		
4. Restorative experience	0.70**	0.67**	0.60**	**0.76**	
5. Spiritual well-being	0.53**	0.47**	0.63**	0.61**	**0.86**
Mean	4.10	4.04	3.73	4.03	5.58
Standard deviation	0.58	0.74	0.86	0.63	0.87

Note: $N = 352$; Diagonal data [in bold] are the square roots of the AVE; $^{**}p < 0.01$.

TABLE 3 | Bias-corrected bootstrap test of mediating effects (Model: TP, FE, SN, RE, SW).

	Estimate (SE)	95% CI (Low)	95% CI (High)
Direct effect			
TP → SW	0.12 (0.10)	−0.08	0.31
Indirect effect			
TP → RE → SW	0.20*** (0.06)	0.11	0.33
TP → FE → SW	−0.05 (0.07)	−0.18	0.09
TP → SN → SW	0.37*** (0.06)	0.26	0.50
TP → FE → RE → SW	0.10** (0.03)	0.05	0.18
TP → SN → RE → SW	0.06* (0.02)	0.02	0.12

Note: $N = 352$; $^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$.
Abbreviations: FE = flow experience; RE = restorative experience; SN = nostalgia; SW = spiritual well-being; TP = therapeutic landscape perception.

TABLE 2 | CFA results.

Models	$\chi 2$	df	$\chi 2$/df	CFI	TLI	RMSEA	SRMR
Baseline five-factor model TP, FE, SN, RE, SW	2476.70	979	2.53	0.90	0.90	0.06	0.05
Four-factor model TP + RE, FE, SN, SW	2852.29	983	2.90	0.84	0.83	0.07	0.06
Three-factor model TP + RE, FE + SN, SW	3234.64	986	3.28	0.81	0.80	0.08	0.07
Two-factor model TP + FE + SN + RE, SW	3830.60	988	3.88	0.76	0.75	0.09	0.07
One-factor model TP + FE + SN + RE + SW	5320.98	989	5.38	0.63	0.61	0.11	0.09

Note: $N = 352$.
Abbreviations: FE = flow experience; RE = restorative experience; SN = nostalgia; SW = spiritual well-being; TP = therapeutic landscape perception.

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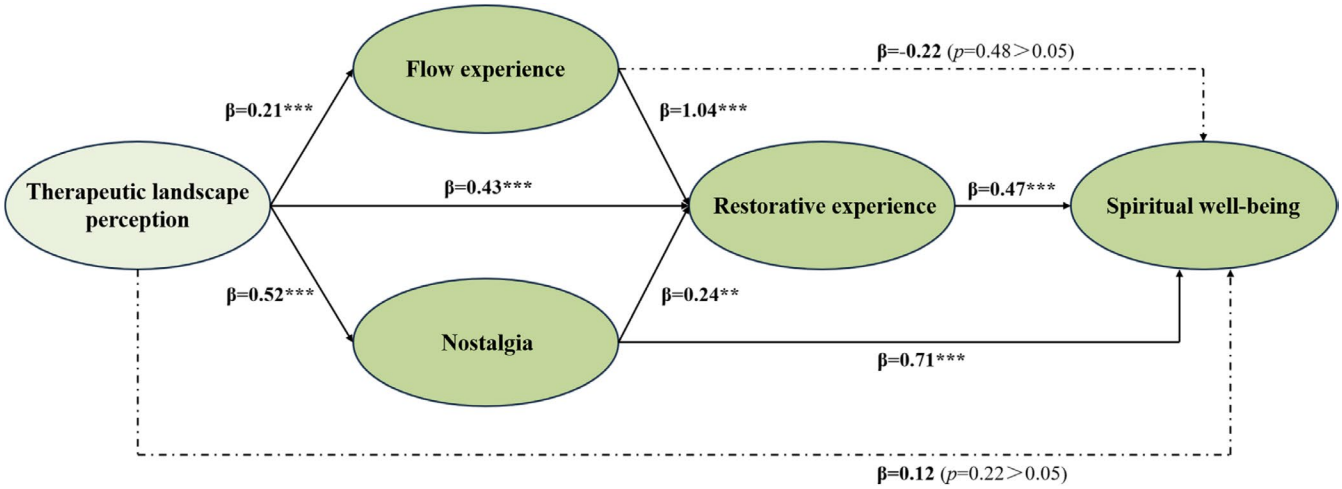


FIGURE 2 | Model with unstandardized coefficient estimates. *Note:* The dotted line indicates a nonsignificant path; $^{**}p < 0.01$; $^{***}p < 0.001$.

5 | Discussion and Conclusions

5.1 | Discussion

This study explores the impact of therapeutic landscapes on tourists, focusing on the mechanisms that shape their spiritual well-being. First, findings indicate that in rural tourism settings, tourists' therapeutic landscape perception positively influences their flow experience, nostalgia, and restorative experience, ultimately enhancing spiritual well-being. These components correspond to the achieving, symbolic, immersive, and social dimensions of Bell et al. (2015) therapeutic experience model. Notably, this study is the first to empirically validate this multidimensional model using quantitative analysis.

Second, this study underscores the mediating role of restorative experience in bridging therapeutic landscape perception and spiritual well-being. Aligning with Bell et al. (2015) concept of immersive therapeutic experience, restorative experience unfolds as tourists escape urban pressures, slow their pace, and reconnect with nature (Kou and Xu 2024; Yuan 2022). By immersing themselves in traditional rural lifestyles, tourists gain mental relief and emotional balance (Meng et al. 2019; Zhai et al. 2025), reinforcing the therapeutic impact of the landscape. This restorative process manifests on multiple levels—physically through nature's harmony, socially via stress-free interactions, and symbolically through cultural narratives—all of which collectively enhance spiritual well-being.

Third, this study investigates how flow experience mediates the relationship between therapeutic landscape perception and spiritual well-being. In Bell et al. (2015) model, flow experience aligns with the achieving dimension. Rural tourism activities such as farming, handicrafts, and outdoor adventures create engaging challenges that foster concentration and enjoyment (King and Church 2013; Liang et al. 2021). As tourists become fully absorbed in these activities, they enter a state of flow, enhancing their sense of achievement and fulfillment (Yang et al. 2023). However, this study finds that flow experience alone does not directly mediate the relationship. Instead, only when paired with restorative experience does it form a chain-mediating effect. This finding suggests that flow experience in a rural setting is insufficient to enhance spiritual well-being on its own. One possible explanation lies in flow's nature, which involves deep task absorption, diminished self-awareness, and altered time perception (DeMatos et al. 2021; Yang et al. 2023). Though flow experience may provide immediate satisfaction (Yang et al. 2023), it is task-oriented rather than reflective, limiting its ability to cultivate meaning or foster personal growth. The loss of self-awareness inherent in flow may also counteract spiritual well-being, which relies on a sense of meaning and purpose (Gedecho and Kim 2024; Han et al. 2024). Moreover, since flow experience hinges on balancing skill and challenge (Jackson and Marsh 1996), it often results in time distortion and detachment from the physical self (Csikszentmihalyi 1990; DeMatos et al. 2021), affecting material rather than spiritual well-being. These findings are consistent with sports research (Spittle and Dillon 2014) and therapeutic landscape studies (Pitt 2014), questioning whether momentary positive states like flow contribute to long-term well-being.

Additionally, nostalgia serves as another key mediator between therapeutic landscape perception and spiritual well-being. In rural tourism, nostalgia fosters symbolic therapeutic experiences by evoking past memories (Kou and Xu 2024; Yuan 2022). The countryside, standing in stark contrast to urban life, reconnects tourists with childhood experiences, family traditions, and ancestral ties, strengthening their emotional bond with rural landscapes (Meng et al. 2019). This nostalgic connection enhances the therapeutic value of rural destinations, fostering a profound sense of love, belonging, and identity, which ultimately contributes to spiritual well-being.

Finally, this study introduces flow experience and nostalgia as dual mediators, representing cognitive and emotional psychological mechanisms, respectively. By incorporating restorative experience into a chain mediation model, it delves deeper into the multiple pathways through which therapeutic landscape perception affects spiritual well-being. This view

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draws from Stienmetz et al. (2021) tourism experience creation framework. Path coefficient analysis reveals a striking contrast: flow experience exerts a more pronounced effect on restorative experience than nostalgia. This resonates with Kaplan (1995) viewpoint of restoration, which emphasizes the role of reducing the reliance on directed attention. In a flow state, individuals are in deep task absorption, reducing the demand on directed attention, replenishing cognitive resources, and enabling its recovery. In contrast, nostalgia, as an emotionally driven recollection, involves reflective engagement with past experiences, which may introduce fluctuations in emotional states and thereby exert a weaker restorative effect. Yet, mediation analysis suggests that nostalgia, as an emotional mechanism, exerts the strongest influence. This underscores its central role in rural tourism research and highlights the profound impact of emotional bonds in shaping tourists' spiritual well-being (Christou et al. 2018; Meng et al. 2019; Yuan 2022).

5.2 | Theoretical Implications

First, this study strengthens the recognition of rural tourism destinations as a distinct category of therapeutic landscapes (Huang and Xu 2018, 2023; Yan and He 2020). While prior research has examined the healing properties of rural settings, many studies rely on small-scale qualitative methods, which, despite their insights, are often constrained by subjectivity and limited generalizability (Zhou, Cheng, and Rosenberg 2023). By adopting a quantitative perspective, this study provides a broader and more structured understanding of how rural therapeutic landscapes enhance tourists' spiritual well-being. Additionally, we contend that therapeutic landscape experiences in rural tourism are not isolated occurrences but rather emerge from a complex transformation process. This argument echoes previous research emphasizing the relational nature of therapeutic landscapes and the intricate interactions between individuals and broader socio-physical environments (Conradson 2005; Huang and Xu 2023; Wang et al. 2018; Wang, Cui, and Xu 2024).

Second, our research positions spiritual well-being as a crucial outcome of engaging with rural therapeutic landscapes. While tourism research has traditionally tied spiritual well-being to religious contexts such as pilgrimages (Han et al. 2024), festivals (Gedecho and Kim 2024), and retreats (Buzinde 2020), nonreligious settings remain underexplored (Gedecho and Kim 2024). Our findings demonstrate that even in secular rural environments, therapeutic landscape perception significantly enhances tourists' spiritual well-being, broadening the scope of spirituality research in tourism. Furthermore, as Bandyopadhyay and Nair (2020) noted, spiritual tourism has yet to receive sustained interdisciplinary focus. By synthesizing perspectives from health geography and tourism research, our study bridges that gap, offering fresh insights into both health and spiritual tourism. Our findings reaffirm the spiritual dimension of therapeutic landscapes in tourism, suggesting that tourism is more than a temporary escape or a liminal experience (Kou and Xu 2024; Zhang and Xu 2019); more significantly, it serves as a catalyst for individuals to have therapeutic experiences, particularly in achieving spiritual well-being. Rather than being passive spaces, rural tourism destinations play an active role in shaping and co-constructing meaning (Huang and Xu 2018, 2023; Kou and Xu 2024; Yan and He 2020). The formation of tourists' spiritual well-being emerges from a relational assemblage of physical, social, and symbolic elements. Therefore, our empirical analysis supports the view of therapeutic landscape research in tourism based on the paradigm of poststructuralism.

Third, this study establishes restorative experience as a mediator between therapeutic landscape perception and spiritual well-being. While prior research has focused on its role in physical and psychological recovery (Hipp et al. 2016; Kaplan and Kaplan 1989; Tang et al. 2024), our study offers a more integrated perspective. We argue that restorative experiences go beyond environmental responses, serving as a pathway to deeper meaning-making. In rural tourism, restoration is not just about stress relief but also fosters a profound sense of spiritual well-being. This nuanced view broadens discussions on restorative experiences in tourism and therapeutic landscapes.

Furthermore, this study questions the assumption that flow experience directly enhances spiritual well-being. While previous research has identified flow as a crucial mechanism in therapeutic experiences (Bell et al. 2015; Pitt 2014), our findings suggest a disconnect between the immediate pleasure of flow and the deeper construction of meaning in rural tourism. Engaging in immersive and challenging activities may generate immediate pleasure (DeMatos et al. 2021), but it does not necessarily lead to long-term spiritual fulfillment. This finding challenges the notion that flow always contributes to deeper existential fulfillment and suggests that the structured and reflective nature of spiritual well-being requires more than transient absorption in an activity.

At the same time, this study identifies nostalgia as a crucial mediator between therapeutic landscape perception and spiritual well-being. This finding aligns with studies in medical contexts, where nostalgia is recognized for its positive emotional impact, its role in fostering continuity, and its contribution to a meaningful life (Wood et al. 2015). In tourism and hospitality, nostalgia has often been linked to experiential value (Chen et al. 2014), place attachment (Jian et al. 2021; Xu and Zhou 2020) and subjective well-being (Karagöz and Ramkissoon 2024). However, our study positions nostalgia as a core mechanism through which tourists derive spiritual fulfillment from rural landscapes. By fostering a temporal connection between past and present, nostalgia enables individuals to reconstruct personal and collective identities within therapeutic landscapes, deepening their spiritual well-being.

Finally, this study elucidates the dual mediating roles of flow experience and nostalgia in linking therapeutic landscape perception, restorative experience, and spiritual well-being. While previous research has focused on how therapeutic landscapes promote relaxation, escape, and recovery (Si et al. 2024; Zhang et al. 2021), few studies have examined the complex prerequisites of restorative experiences. Our findings address this gap by considering both cognitive (flow experience) and emotional (nostalgia) pathways. The results validate the dual-channel experience creation framework (Stienmetz et al. 2021) and underscore the transformative role of restorative experience in

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shaping spiritual well-being. We argue that restorative experiences are not just about resource replenishment; rather, they are also about meaning reconstruction, allowing tourists to transform flow and nostalgia into deeper spiritual well-being.

5.3 | Practical Implications

This study offers key recommendations for rural tourism destination management. First, restorative experience plays a vital role in linking tourists' therapeutic landscape perception to spiritual well-being. Destinations should prioritize relaxation and mental rejuvenation by preserving natural scenery, minimizing urban intrusions, and promoting slow-paced tourism (Kou and Xu 2024). Second, nostalgia is a powerful emotional driver in rural tourism. Managers can tap into this by emphasizing local heritage, traditional culture, and a strong sense of place (Meng et al. 2019). Storytelling, interactive exhibits, and reminiscent experiences (e.g., heritage homestays or traditional handicrafts) can deepen tourists' emotional connection to the destination. Third, while flow experience alone does not directly enhance spiritual well-being, its impact grows when paired with restorative experience. This highlights the need for rural destinations to offer engaging yet calming activities, such as guided farming, meditation retreats, or outdoor yoga (Huang and Xu 2018; Liang et al. 2021). Ultimately, rural tourism stakeholders should adopt a holistic approach that integrates these psychological mechanisms. By integrating natural, cultural, and artistic elements, they can enhance rural tourism's therapeutic value, creating a deeply enriching and well-being-oriented experience.

5.4 | Limitations and Future Research

Several limitations should be acknowledged. First, this study relies on survey data from 352 tourists at Chinese rural tourism destinations. While the sample provides valuable insights, it may not generalize to other cultural or geographical contexts. Future research should explore rural destinations in other countries to examine potential cultural variations in therapeutic landscape perception and its psychological effects. Second, this study uses a three-wave questionnaire survey, which helps reduce CMB through procedural remedies (Podsakoff et al. 2003). However, self-reported data are still subject to social desirability and recall biases. Experimental methods could help establish stronger causal relationships in future studies. Third, perceptions of therapeutic landscapes change over time. These shifts are shaped not only by changes in the landscape itself and individuals' life experiences (Wood et al. 2015) but also by the duration of their interaction with the environment (Yin et al. 2020). To capture these evolving experiences, future studies could adopt longitudinal approaches, such as extended tracking periods (e.g., assessing experiences during a trip, the fourth week after a trip, and the eighth week after a trip; Yu et al. 2021) or daily diary methods (Cui 2023; Hanel et al. 2024).

Author Contributions

Yinger Xu: conceptualization, data curation, formal analysis, writing – original draft, writing – review and editing. **Yang Yang:** conceptualization, methodology, investigation, writing – original draft, writing – review and editing. **Haizhou Zhang:** validation, writing – original draft. **Xing Zhou:** conceptualization, funding acquisition, supervision.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Research data are not shared.

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Appendix A

Items

Therapeutic Landscape Perception

TP-1: There are comfortable blue sky and white clouds.

TP-2: There are comfortable lake/sea and lakeside/beach.

TP-3: There are comfortable green plants.

TP-4: There are places for people to exercise and have fun.

TP-5: This place has healthy ingredients.

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TP-6: There are physical and mental healing scenes.

TP-7: I could get to know residents.

TP-8: I could get along well with others.

TP-9: I could exchange life experiences with others.

TP-10: To be here at this time of year is a fitting healing journey.

TP-11: This is one of the more representative rural therapeutic places around.

TP-12: This is a great place for those who want to come to the countryside to heal.

Flow Experience

FE-1: I experienced flow during this travel.

FE-2: In general, how frequently would you say you have experienced "flow" when traveling.

FE-3: Most of the time when I travel, I feel that I am in flow.

Nostalgia

SN-1: The scenery in the countryside made me feel nostalgic.

SN-2: The smell of the countryside makes me feel nostalgic.

SN-3: The sounds in the countryside make me feel nostalgic.

SN-4: This place makes me think of precious times in the past.

SN-5: This place makes me want to go back to good old days.

SN-6: In the countryside, I can feel more nostalgia for my childhood than I have in the past month.

SN-7: This place makes me think of unforgettable family and friends.

Restorative Experience

RE-1: It is a confusing place.

RE-2: There is a great deal of distraction.

RE-3: It is chaotic here.

RE-4: Being there suits my personality.

RE-5: The things I like to do can be done at the place.

RE-6: I have a sense of oneness with the place I visited.

RE-7: I felt removed from my daily routines.

RE-8: A visit to this place is an escape experience.

RE-9: Spending time here gives me a good break from my daily stress.

RE-10: My attention is drawn to many interesting things about this place.

RE-11: I found the destination fascinating.

RE-12: There was much to explore and discover at the destination.

Spiritual Well-Being

SW-1: I have developed a very good self-awareness.

SW-2: I could develop joy in life.

SW-3: I have found my meaning of life.

SW-4: I have developed connection with nature.

SW-5: I could develop awe at breathtaking view.

SW-6: I have developed harmony with the environment.

SW-7: I could develop forgiveness towards others.

SW-8: I could develop kindness towards other people.

SW-9: I could develop a love for other people.

SW-10: I could develop oneness with the Divine/God/universe.

SW-11: I could develop peace with the Divine/God/universe.

SW-12: I could develop worship with the spirit and law of the Divine/God/universe.

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Appendix B

Factor Loadings, Cronbach's α, Composite Reliability, and Average Variance Extracted

Variables	Items	Loading	Cronbach's α	CR	AVE	Variables	Items	Loading	Cronbach's α	CR	AVE
Therapeutic landscape perception	TP-1	0.51	0.88	0.93	0.53	Restorative experience	RE-1	0.78	0.92	0.94	0.58
	TP-2	0.52					RE-2	0.82			
	TP-3	0.76					RE-3	0.77			
	TP-4	0.66					RE-4	0.75			
	TP-5	0.79					RE-5	0.80			
	TP-6	0.75					RE-6	0.69			
	TP-7	0.77					RE-7	0.74			
	TP-8	0.80					RE-8	0.70			
	TP-9	0.71					RE-9	0.77			
	TP-10	0.74					RE-10	0.81			
	TP-11	0.85					RE-11	0.81			
	TP-12	0.81					RE-12	0.70			
Flow experience	FE-1	0.82	0.86	0.86	0.68	Spiritual well-being	SW-1	0.83	0.96	0.97	0.74
	FE-2	0.85					SW-2	0.83			
	FE-3	0.80					SW-3	0.88			
Nostalgia	SN-1	0.77	0.93	0.93	0.65		SW-4	0.86			
	SN-2	0.82					SW-5	0.82			
	SN-3	0.81					SW-6	0.84			
	SN-4	0.85					SW-7	0.88			
	SN-5	0.85					SW-8	0.88			
	SN-6	0.84					SW-9	0.88			
	SN-7	0.71					SW-10	0.91			
							SW-11	0.89			
							SW-12	0.84			

Note: $N = 352$.
Abbreviations: AVE = average variance extracted; CR = composite reliability.

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